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EB likelihood function:

- $W\{[Fe/H], t, M_1, M_2, e\sin\omega, e\cos\omega\}$: Posterior distribution from Windemuth et al (2019)

- $\mathcal{E}_e(e_{env})$: envelope eccentricity distribution @ P_{orb}
- e_{max}: Calculated max eccentricity from orbital evol

$$\mathcal{L} = \mathcal{L}_{max}(e, e_{max})\, W\{[\tfrac{Fe}{H}], t, M_1, M_2, e\sin\omega, e\cos\omega\}$$
$$\pi(Q)\,\pi(\omega)\,\pi(e)$$

with:

$$\mathcal{L}_{max} \equiv \begin{cases} 1 & \text{if } \; e < e_{max} < e_{env} \\ 0 & \text{otherwise} \end{cases}$$

W already includes priors on binary parameters. In particular, $\pi(e\sin\omega) = \pi(e\cos\omega) = \mathcal{U}(-1, 1)$. This biases eccentricity away from zero!
So we take:

$$\pi(e) = \frac{1}{e}$$

Marginalize over ω by replacing ~~samples W19~~
$e\sin\omega,\ e\cos\omega \rightarrow \sqrt{e^2(\sin^2\omega + \cos^2\omega)}$ in W19 samples. $W \to W'$

Marginalize over e_{env}:

$$\mathcal{L} = H(e_{max} - e)\, W'(\ldots)\, \frac{1}{e}\, \pi(Q) \left[1 - \hat{\mathcal{E}}_e(e_{max})\right]$$

$$W' = W'\{[\tfrac{Fe}{H}], t, M_1, M_2, e\}$$

To marginalise over e: ②

$$\pi(Q)\left[1-\hat{\mathcal{E}}_e(e_{env})\right]\int_0^1 de\; H(e_{max}-e)\, W'\left\{\left[\tfrac{Fe}{H}\right], t, M_1, M_2, e\right\}\frac{1}{e}$$

$$= \pi(Q)\left[1-\hat{\mathcal{E}}_e(e_{max})\right]\int_0^{e_{max}} W'\left\{\left[\tfrac{Fe}{H}\right], t, M_1, M_2, e\right\}\frac{de}{e}$$

!

But $e_{max} = e_{max}\left(\left[\tfrac{Fe}{H}\right], t, M_1, M_2\right)$

$\cancel{\mathcal{L} = \pi(Q)\int_0^{e_{env}}}$

Let $\lambda(x, M_1, M_2, t, [\tfrac{Fe}{H}]) \equiv \int_0^x W'\left\{\left[\tfrac{Fe}{H}\right], t, M_1, M_2, e\right\}\frac{de}{e}$

Also let $\mathcal{E}_e \rightsquigarrow \delta$

$$\Rightarrow \mathcal{L} = \pi(Q)\,\lambda\left(e_{env}, \left[\tfrac{Fe}{H}\right], t, M_1, M_2\right) \times \frac{\lambda\left\{e_{max}\left(\left[\tfrac{Fe}{H}\right], t, M_1, M_2\right), \left[\tfrac{Fe}{H}\right], t, M_1, M_2\right\}}{\lambda\left(e_{env}, \left[\tfrac{Fe}{H}\right], t, M_1, M_2\right)}$$

Top line: priors

Bottom line: MCMC likelihood

Approximate W' using KDE:

$$W' = \sum_s k\left(\frac{[\frac{F_x}{H}] - [\frac{F_x}{H}]_s}{w_{[\frac{F_x}{H}]}}\right) k\left(\frac{t - t_s}{w_t}\right) k\left(\frac{(\mu_1,\mu_2) - (\mu_1,\mu_2)_s}{w_\mu}\right)$$
$$k\left(\frac{\frac{\mu_2}{\mu_1} - \left(\frac{\mu_2}{\mu_1}\right)_s}{w_q}\right) k\left(\frac{e - e_s}{w_e}\right)$$

$$\Rightarrow \lambda\left(e_{curv}, [\tfrac{F_x}{H}], t, \mu_1, \mu_2\right) = \sum_s k_{[\frac{F_x}{H}]}\, k_t\, k_{\mu_1\mu_2}\, k_{\frac{\mu_2}{\mu_1}} \times$$
$$\times \int_0^{e_{curv}} k\left(\frac{e - e_s}{w_e}\right) \frac{de}{e}$$

Where $k_? = k\left(\frac{? - ?_s}{w_?}\right)$

Implementation

(4)

For each sample, calculate $E_s \equiv \int_0^{e_{max}} k\left(\frac{e-e_s}{w_e}\right)\frac{de}{e}$

and replace e column with these values.

To sample $\left[\frac{Fe}{H}\right]$, give $u_{[\frac{Fe}{H}]} \sim \mathcal{U}(0,1)$

Pre-compute ~~and store it~~:

$$\hat{\varphi}\left(\left[\frac{Fe}{H}\right]\right) \equiv \int_{-\infty}^{\infty} dt \int_{-\infty}^{\infty} d(M_1+M_2) \int_{-\infty}^{\infty} d\left(\frac{M_2}{M_1}\right) \int_{-\infty}^{[\frac{Fe}{H}]} d\left[\frac{Fe}{H}\right]' \sum_s k_{t,s}\, k_{M_1+M_2,s}\, k_{\frac{M_2}{M_1},s}\, k_{[\frac{Fe}{H}]',s}\, E_s$$

since $\int_{-\infty}^{\infty} dt\, k_{t,s} = 1 \ \forall s$ etc.

$$\Rightarrow \hat{\varphi}\left(\left[\frac{Fe}{H}\right]\right) = \sum_s E_s \int_{-\infty}^{\infty} d\left[\frac{Fe}{H}\right]' k\left(\frac{\left[\frac{Fe}{H}\right]' - \left[\frac{Fe}{H}\right]_s}{w_{[\frac{Fe}{H}]}}\right)$$

$$\left[\frac{Fe}{H}\right]_N = \hat{\varphi}^{-1}\left(u_{[\frac{Fe}{H}]}\right)$$ so we store $\hat{\varphi}^{-1}$

Similarly, pre-compute & store $\hat{\tau}^{-1}\left(\left[\frac{Fe}{H}\right], u_t\right)$

Where

$$\hat{\tau}\left(\left[\frac{Fe}{H}\right], t\right) \equiv \sum_s E_s\, k\left(\frac{\left[\frac{Fe}{H}\right] - \left[\frac{Fe}{H}\right]_s}{w_{[\frac{Fe}{H}]}}\right) \int_{-\infty}^{t} k\left(\frac{t'-t_s}{w_t}\right) dt'$$

M_1+M_2 sampling can be stored as 3-D function or computed in real time.

$\frac{M_2}{M_1}$ sampling is impractical to store!

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